

MCR3M - M1C2

Agenda:

1. Finish Example from Wednesday
2. Minds-On: Class Starter
3. Introduction to Portfolio Assignment
4. Open Office

Learning Goal:

I will understand the first story requirement
for the assignment

Linear vs. Quadratic Functions (From Last Class)

How can you tell if a relationship is linear or quadratic?

The language used depends on the representation:

Algebraic (an equation): Check for the highest **degree**

Graphical: Just look! Is it straight or parabolic?

Table of Values: Calculate First & Second Differences

Ex. 4: Identify each relation as linear or quadratic:

a) $f(x) = 4x - 1$

b)

c)

Degree is 1

⇒ Linear

($y = mx + b$)

x	y	FD	SD
0	0		
1	3	$3 - 0 = 3$	2
2	8	$8 - 3 = 5$	2
3	15	$15 - 8 = 7$	2
4	24	$24 - 15 = 9$	↓

Not Linear Quadratic

Degree (of a polynomial):

The degree of a polynomial is the highest exponent of the variable(s) in any single term (monomial) of the expression

Are First Differences constant? Linear

Are Second Differences constant? Quadratic

Formative from Last Class

Q7: Evaluate $f(2)$ if $f(x) = x^2 - 3x + 4$. Show your steps (2-3 lines)

 Sarah Everett

$$f(2) = 2^2 - 3(2) + 4$$

$$f(2) = 4 - 6 + 4$$

$$f(2) = 2$$

Q10: A relation is given by the set of ordered pairs: $(-2, 1), (-1, 3), (0, 5), (1, 7), (2, 9)$

1. Determine whether this relation is a function.
2. Decide whether the relationship is more consistent with a linear function or a quadratic function, and justify your choice using a numerical feature (e.g. first or second differences).
3. Using your conclusion, write a possible rule for the relation and use it to find the output when the input is 3.

 Serena Douglas

1. Yes, this relation is a function because with each specific input (x-value) there is a unique output (y-value).
2. This is a linear function. I know this because when calculating the first difference the answer was consistently 2, making it linear.
3. A possible rule for this linear function is $f(x) = 2x + 5$
 $f(3) = 2(3) + 5$
 $= 6 + 5$
 $= 11$

Minds-On: Class Starter

M1C2 Formative Question 1

Evaluate the quadratic function $h(x) = -2(x - 3)^2 + 1$ at $x = 5$.

Show all steps and write your final answer using correct function notation (for example, $h(5)$).

Type a response



Portfolio Assignment: Mathematical Stories

What? An assignment that replaces traditional homework assignments with a curated collection of mathematical stories.

The goal: To show how your understanding develops over time: through struggle, error analysis, explanation, and extension. You will build this portfolio throughout the module (approximately two weeks) and then discuss it in a one-on-one conference with me.

To Do: Use at least three (3) problems from this module's homework (the assigned textbook questions) to create three mathematical stories

A story is a short, focused narrative showing how your understanding of a specific type of math problem or concept developed, supported by concrete evidence (questions, work, checks, AI transcripts, and reflection).

Story 1:



Purpose: Using at least one (1) textbook question, show how your understanding changed.

Required Components:

- **Initial Attempt (Struggle)**
 - A type of problem you initially found difficult
 - Include an image of the question as well as your original answer and work (consider one from class formative work, but a problem from the web or the textbook works too)
- **Revised Understanding (Breakthrough)**
 - A *new* question or sequence of questions that tests the **same underlying concept** (not just similar difficulty)
 - Include an image and your answer and work for each
- **Reflection**
 - What did I misunderstand at first?
 - What helped me see it differently?
 - What do I now understand about this concept that I didn't before?

Cheat Code: If you struggle early in this module, that's not a problem — that's actually Story 1 waiting to happen!

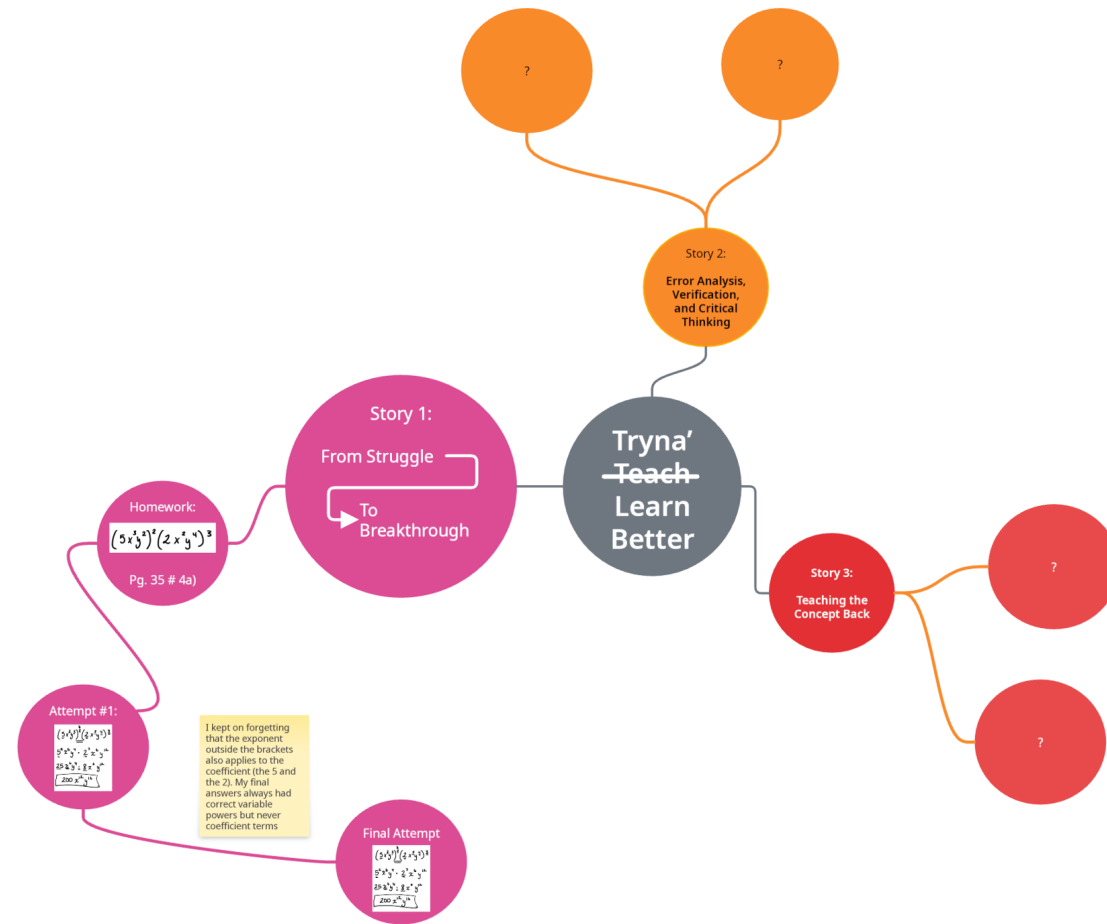
Start with a formative question from class that you honestly tried. After you've used my feedback, keep that feedback in mind while jumping into similar homework questions. Do this for each lesson and you'll eventually see at least one of the process click. When you do, notice what changes in how you think.

If you can tell the story of what didn't make sense at first and what clicked later, that's an authentic growth story — and that's exactly what I'm looking for.



Example Portfolio Template

My Module 1 Story





Remaining Time: **Open Office**

I will try to leave **at least 30 minutes every Friday for Open Office** (some days it will be the entire 75 minutes)

Why attend Open Office? You can...

1. *get a more detailed explanation of topics you may have been confused about in class*
2. *ask any questions you have about homework (textbook) problems*
3. *get direction about your assignment stories*
4. *work in the background in case other students ask questions you find interesting*
5. *discuss how math extends beyond the typical "math problems" seen in class*